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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B. Tech

Semester: 7th

Name of the Paper Transportation Engineering II

Paper Code: TCE -703

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Write short on: 1. Wind resistance. 2. Resistance due to track profile. 3. Starting and accelerating resistance. 4. Train resistance	CO1
(b)	What are the possible causes of creep? What are the effects of creep?	
(c)	Find out the steepest gradient on a straight track using the following data, for a train having 20 wagons. Weight of each wagon =18 tones ,Rolling resistance of wagon = 2.5 kg/ ton, Speed of the train =50km/hr. Weight of locomotive =120 tones ,tractive effort of locomotive =12 tones rolling resistance of locomotive=3.5kg/tones ,	
Q2	(20 marks)	
(a)	Define composite sleeper index. Using a sleeper density of M+5, find out the Number of sleepers required for a constructing a railway track 640 meters long (B.G Track)	CO2
(b)	Discuss different type of rail joints with the help of neat sketches and give their merits and limitations.	
(c)	Draw a typical cross section of a permanent way .discuss the brief the basic function of a various components of a railway track	
Q3	(20 marks)	
(a)	Explain the necessity of widening of gauge on curves. And find out the extra widening on B.G. track of a circular curve of radius 450 meter , using the following data: 1. Length of rigid wheel base = 6 meter 2. Depth of the flange below the top of the rail=4 cm 3. Diameter of wheel = 1.3 meter	CO3
(b)	What is can't deficiency? If branch curve of 8° is diverge for main curve of 5° is an opposite direction in a layout of B.G .track compute the super elevation & speed of branch line , if the maximum speed permitted on main line is 45km/hr.	
(c)	What are the objects of providing transition curves on railway? Calculate the cant deficiency & permissible speed for 4° curve of B.G. track. Also find the equilibrium cant & length of transition curve.	

Q4	(20 marks)	
(a)	What are the objects of signaling? Explain the different types of signals used in station yards.	CO4
(b)	If a cross over is formed between two turnouts in for 1 in 12 at one end and 1 in 16 at another end, with a waved portion between two turnouts. Centre to center distance in between tracks is 6.2 meters .compute overall length of cross over.	
(c)	What are the functions of points and crossing in railway track layout? Discuss the various types of switches.	
Q5	(20 marks)	
(a)	Briefly explain drilling of shaft and drilling equipment.	CO5
(b)	What is the necessity of railway tunnels? Draw a sketch to illustrate a single track railway tunnel.	
(c)	Describe the term hauling of muck and ventilation & lightning of tunnels.	